Class-6 Chapter 6: Material Around Us— Complete Solution NCERT New Syllabus

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A. Choose the correct answer

1. Which of the following materials is hard?

(a) Chalk
(b) Sponge
(c) Gold
(d) Cotton

Answer: (c) Gold. Gold is a hard material compared with chalk, sponge and cotton.

2. Which of the following properties does wood have?

(a) Hard
(b) Non-lustrous
(c) Insulator of heat
(d) All of these

Answer: (d) All of these.

3. Which of the following is miscible with water?

(a) Vinegar
(b) Cooking oil
(c) Kerosene oil
(d) Petrol

Answer: (a) Vinegar. Vinegar mixes completely with water.

4. Which of the following solids are insoluble in water?

(a) Chalk powder
(b) Sand
(c) Humus
(d) All of these

Answer: (d) All of these.

5. Which of the following units is used to measure mass?

(a) L
(b) mL
(c) kg
(d) m

Answer: (c) kg. Kilogram is a standard unit of mass.

B. Write True or False. Also, correct the incorrect statements.

1. Materials that do not allow heat to flow through them are called conductors.

Answer: False. Materials that do not allow heat to pass through them easily are called insulators of heat.

2. Light does not pass through opaque materials.

Answer: True.

3. Liquids take the shape of the container in which they are stored.

Answer: True.

4. Smooth surfaces are full of bumps and ridges.

Answer: False. Rough surfaces have bumps and ridges. Smooth surfaces do not have many visible bumps and ridges.

5. Oil and water are miscible with one another.

Answer: False. Oil and water are immiscible. They do not mix with each other.

C. Match the columns

Column A	Correct Match
1. Sugar	(d) Water soluble
2. Plastic	(c) Insulator of electricity
3. Nitrogen	(a) Gas
4. Aluminium	(e) Good conductor of heat
5. Frosted glass	(b) Translucent

D. Give two properties of each material

1. Iron

- It is hard and strong.
- It is a good conductor of heat and electricity.

2. Wood

- It is generally hard and non-lustrous.
- It is a poor conductor of heat and electricity.

3. Glass

- It is generally transparent.
- It is hard but brittle.

4. Plastic

- It is generally light in weight.
- It is a poor conductor of heat and electricity.

5. Air

- It is a gas.
- It has no fixed shape or volume.

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E. Give one word for the following

1. Materials that do not allow light to pass through them.

Answer: Opaque materials.

2. Materials that allow light to pass through them completely.

Answer: Transparent materials.

3. Materials that allow light to pass through them partially.

Answer: Translucent materials.

4. Materials that allow electricity to pass through them.

Answer: Conductors of electricity.

5. Materials that do not allow heat to pass through them.

Answer: Insulators of heat.

F. Give two examples of each

1. Smooth materials

- Glass
- Polished marble

2. Non-lustrous materials

- Wood
- Chalk

3. Transparent materials

- Clear glass
- Clean water

4. Solids

- Stone
- Wood

5. Hard materials

- Iron
- Gold

Crisp and Accurate – Short Answer Questions

A. Define the following terms

1. What is solubility?

Answer: Solubility is the ability of a substance to dissolve in a liquid. For example, sugar dissolves in water.

2. What is density?

Answer: Density is the mass of a substance in a given volume. It tells us how closely the matter is packed in a substance.

3. What is transparency?

Answer: Transparency is the property of a material that allows light to pass through it. For example, clear glass is transparent.

4. What are miscible liquids?

Answer: Miscible liquids are liquids that mix completely with each other to form a uniform mixture. For example, water and vinegar.

5. What is classification?

Answer: Classification is the process of grouping things according to their similar properties.

B. Answer the following questions briefly

1. Oxygen that we use for breathing is an example of which state of matter?

Answer: Oxygen is a gas. Therefore, it is an example of the gaseous state of matter.

2. Name two substances that are soluble and two substances that are insoluble in water.

Answer:

- Soluble substances: Sugar and salt.
- Insoluble substances: Sand and chalk powder.

3. How is wood different from glass?

Answer: Wood and glass have different properties.

Wood	Glass
It is generally opaque.	It is generally transparent.
It is non-lustrous.	It can be lustrous.
It is a poor conductor of heat.	It is also a poor conductor of heat.
It is generally not brittle.	It is brittle and can break easily.

4. Why does cooking oil float on water?

Answer: Cooking oil floats on water because it is less dense than water. Oil and water do not mix with each other.

5. What do you mean by conductors of heat? Give two examples.

Answer: Conductors of heat are materials that allow heat to pass through them easily.

Examples:

1. Iron
2. Aluminium

Think and Explain – Long Answer Questions

1. Explain how materials can be classified on the basis of transparency.

Answer: Materials can be classified into three groups on the basis of how much light passes through them.

1. Transparent Materials

These materials allow light to pass through them completely. We can see objects clearly through them.

Examples:

- Clear glass
- Clean water

2. Translucent Materials

These materials allow only some light to pass through them. We cannot see objects clearly through them.

Examples:

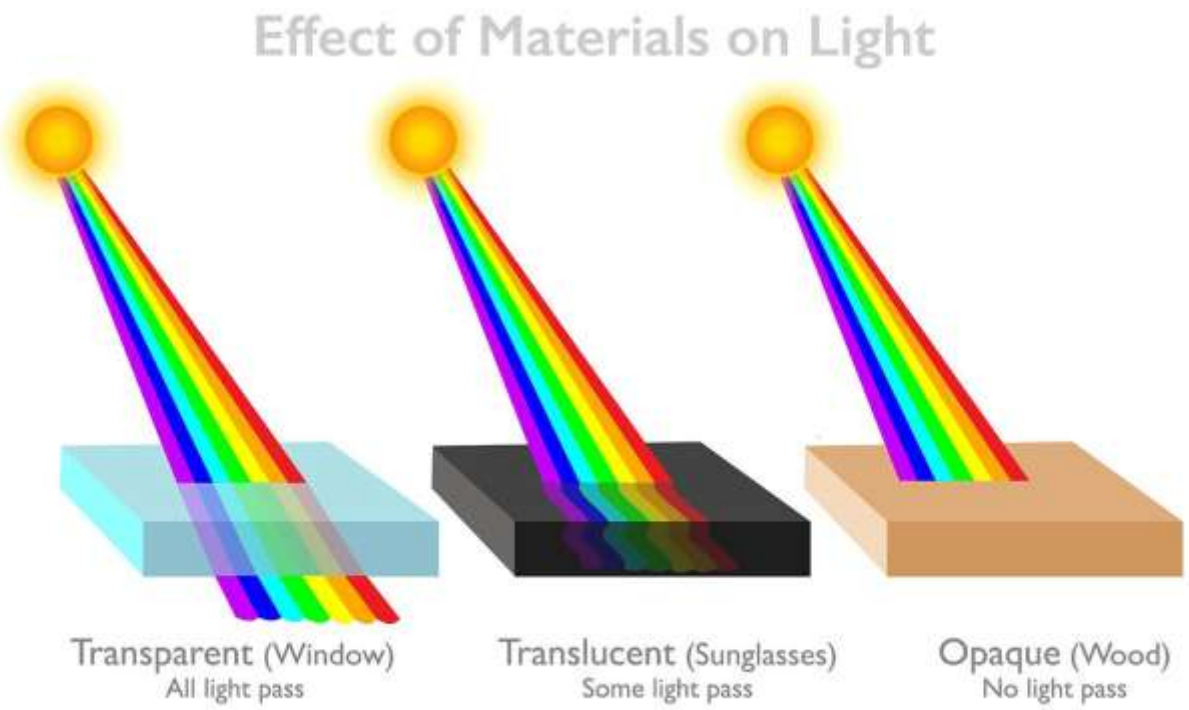
- Frosted glass
- Butter paper

3. Opaque Materials

These materials do not allow light to pass through them. We cannot see through them.

Examples:

- Wood
- Stone



2. How will you classify materials on the basis of their appearance and texture?

Answer: Materials can be classified on the basis of their appearance and texture in the following ways.

A. On the basis of appearance

1. Lustrous Materials: Materials that have a shiny surface are called lustrous materials. Examples: Gold and silver.
2. Non-lustrous Materials: Materials that do not have a shiny surface are called non-lustrous materials. Examples: Wood and chalk.

B. On the basis of texture

1. Smooth Materials: Materials that have an even surface are called smooth materials. Examples: Glass and polished marble.
2. Rough Materials: Materials that have an uneven surface with bumps and ridges are called rough materials. Examples: Sandpaper and tree bark.



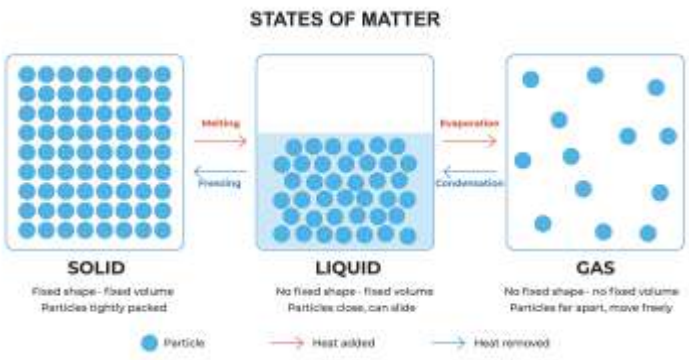
3. Differentiate between the three states of matter. Also give two examples of each type.

Answer: Matter is commonly found in three states: solid, liquid and gas.

Property	Solids	Liquids	Gases
Shape	Have a fixed shape	Take the shape of the container	Have no fixed shape
Volume	Have a fixed volume	Have a fixed volume	Have no fixed volume
Flow	Do not flow	Can flow	Can flow easily
Arrangement of particles	Particles are closely packed	Particles are less closely packed	Particles are far apart

Examples:

- Solids: Stone and wood.
- Liquids: Water and cooking oil.
- Gases: Oxygen and nitrogen.



4. Give the classification of materials on the basis of solubility.

Answer: Materials can be classified into two groups on the basis of their solubility in water.

1. Soluble Materials

Materials that dissolve in water are called soluble materials.

Examples:

- Sugar
- Salt

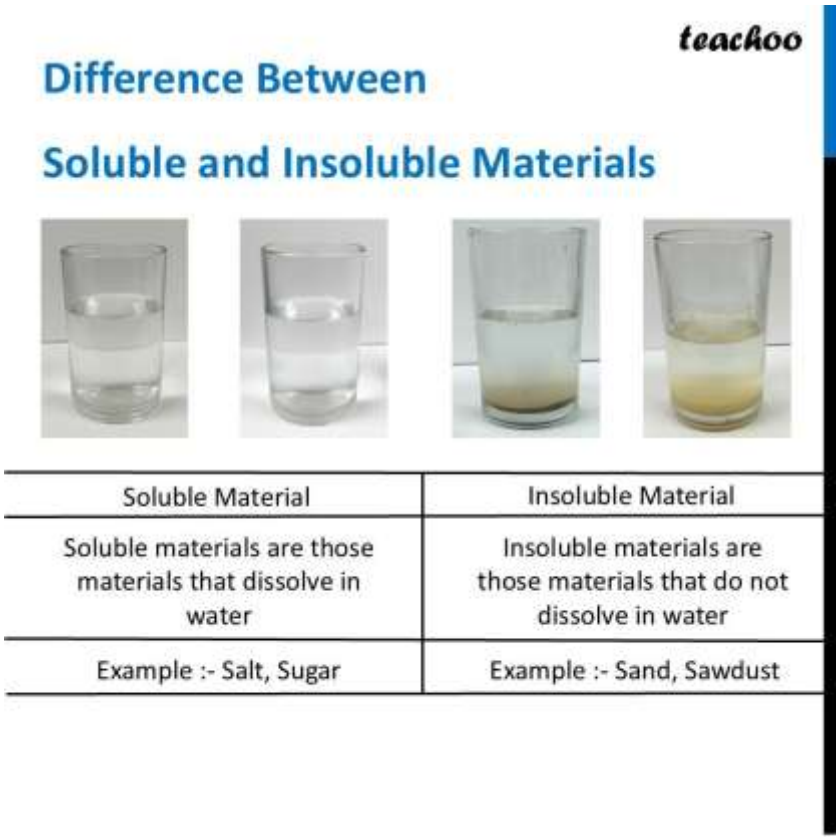
2. Insoluble Materials

Materials that do not dissolve in water are called insoluble materials.

Examples:

- Sand
- Chalk powder

Conclusion: Some materials dissolve in water, while others do not. This property is called solubility.



5. State the general properties of metals and plastics.

Answer:

A. General Properties of Metals

1. Metals are generally hard and strong.
2. They are usually lustrous or shiny.
3. They are generally good conductors of heat.
4. They are generally good conductors of electricity.
5. Most metals are solids at room temperature.
6. Many metals can be beaten into thin sheets and drawn into wires.

Examples: Iron, copper, aluminium and gold.

B. General Properties of Plastics

1. Plastics are generally light in weight.
2. They can be moulded into different shapes.
3. Most plastics are poor conductors of heat.
4. Most plastics are poor conductors of electricity.
5. They do not rust like iron.
6. They are available in many colours and forms.

Examples: Plastic bottles, buckets, toys and pipes.

Metals	Plastics
Generally hard and strong.	Properties vary; many are light and flexible.
Usually shiny.	May be shiny or dull.
Generally conduct heat and electricity.	Generally insulate against heat and electricity.
Many can be beaten into sheets.	Can be moulded into different shapes.



Quick Revision – Important Points

1. Materials are the substances from which objects are made.
2. Materials can be classified according to their properties.
3. Gold and iron are examples of metals.
4. Wood and plastic are generally poor conductors of heat.
5. Transparent materials allow light to pass through completely.
6. Translucent materials allow light to pass through partially.
7. Opaque materials do not allow light to pass through.
8. Lustrous materials have shiny surfaces.
9. Rough surfaces have bumps and ridges.
10. Soluble substances dissolve in water.
11. Insoluble substances do not dissolve in water.
12. Oil and water are immiscible liquids.
13. Solids have a fixed shape and volume.
14. Liquids have a fixed volume but no fixed shape.

15. Gases have neither a fixed shape nor a fixed volume.
16. Kilogram (kg) is a unit of mass.
17. Metals are generally good conductors of heat and electricity.
18. Plastics are generally poor conductors of heat and electricity.

Remember: Different materials have different properties. We choose materials for making objects according to their properties and uses.

These answers are based on the exercise questions shown on pages 88–89 of your uploaded PDF.

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